To: The Editor-in-Chief
Measurement and Control

From:
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Re: Submission of manuscript
"*Real-Time Robust HOCBF Safety Filtering for Supercritical Power Plant Control under Model Mismatch*"

Dear Editor,

We submit the enclosed manuscript for consideration at Measurement and Control. The paper presents an industrial measurement-and-control safety filtering framework that preserves pressure, enthalpy, and power output constraints during supercritical power plant operation under model mismatch.

Fit with Measurement and Control. The journal's scope encompasses measurement, control, and automation in industrial applications. Our work directly addresses this scope along four dimensions:

1. **Industrial measurement and control relevance.** Supercritical boiler-turbine units require continuous monitoring and constraint-preserving control of steam pressure, separator enthalpy, and power output. Model mismatch from coal quality variation, equipment aging, and unmodeled operating disturbances is a daily operational reality. Our safety filter provides a principled, real-time mechanism for maintaining these industrial operating constraints.
2. **Real-time safety filtering.** The method functions as an online safety projection layer that can be superimposed after any upstream controller—LQR, MPC, or a learned policy—without modifying the existing control architecture. Per-step safety-filter pipeline latency is $\sim$25 ms (GPU, JIT-compiled), including the PPO forward pass, GP inference, and QP solve, and is compatible with typical distributed control system (DCS) sampling rates.
3. **Constraint-preserving operation.** Under model mismatch, nominal safety filters may certify unsafe control actions as safe. Our framework transforms catastrophic filter failure ($> 95\%$ constraint violation) to no observed violations across six static perturbation scenarios on a 5th-order supercritical benchmark.
4. **Model-mismatch-resilient control implementation.** The method quantifies and certifies model uncertainty through Gaussian process residual learning with compositional uncertainty propagation, supporting a high-probability forward-invariance certificate under fixed calibrated-GP assumptions, while remaining compatible with standard industrial control architectures.

Key contributions.

1. Compositional GP uncertainty propagation through the HOCBF ψ-chain, yielding a per-constraint robustness margin with a high-probability forward-invariance certificate (Theorem 1).
2. Controller-agnostic deployment validated with both LQR and PPO upstream controllers: identical safety outcomes across controller types.
3. Validation on a 5th-order supercritical boiler-turbine benchmark with Φ-scaled nonlinear dynamics: nominal HOCBF fails ($> 95\%$ violation); Robust HOCBF achieves no observed violations with per-step latency $\sim$25 ms, approximately ten times faster than the implemented SLSQP-based NMPC baseline.

Conflicts of interest. The authors declare no competing interests. The manuscript has not been published elsewhere and is not under simultaneous consideration by any other journal.

Sincerely,

Zhuang Shao (on behalf of all authors)
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